14 A Bibliometric Analysis on Application of Artificial Intelligence (AI) in Workforce Management

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14.1 INTRODUCTION

The basic reason for data mining is to remove significant data from information that is received and transform it into a beneficial example or pattern that can be used further. Subsequently, there is a developing interest in utilizing information mining strategies in the workforce on the viability and standard of work and learning. In any case, to have a significant impact, the accumulated instructive information should, at least, be grasped and transformed into helpful information (Khang et al., 2023a).

The term "artificial intelligence" (AI) alludes to the utilization of information mining procedures in instructive information. Essentially all instructive exercises are upheld by the broad utilization of data and correspondence innovation in instructive foundations and preparing offices, which has additionally brought about

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a gigantic amount of collected information (Snehal, Babasaheb & Khang, 2023). These frameworks accumulate information, as well as data about the upheld processes, through careful logging.

Information gives helpful subtleties on how employees/workforce were assessed, how they utilized learning assets, how they were associated with each other, the means they took, and how they were by and large engaged with the working process.

Likewise, the utilization of mixed working like in online mode systems gives helpful information with respect to work practices at different levels of granularity (Babasaheb et al., 2023b).

Machine learning (ML) is a comparable field of study that generally uses information mining procedures. By inspecting the information that has been gathered, ML has the ability to recognize and characterize the properties of learning methods.

As per Jain et al.'s (2023) study, learning/content management systems (LMS/CMS) are being utilized by instructive foundations to gather information connected to upgrading the instructive process for their workforce in order to gain experiences because of the boundless utilization of AI and ML (Rana et al., 2021).

In LMS logs, evaluation information, and different types of instructive information have all been actually utilized information mining methods, including affiliation investigation and order (Gupta et al., 2023).

In spite of the fact that AI might be utilized to more easily comprehend instructive processes and deal with substitute answers for cooperation issues during those processes, most conventional AI approaches put a greater amount of emphasis on the result than the process (Khang et al., 2023b). In light of the information, AI offers a powerful strategy for examining functional processes (Khang, Rani & Sivaraman, 2022b).

14.2 PROBLEM STATEMENT

The application of artificial intelligence in workforce management is a growing technique for overcoming new barriers. This empowers the revelation of interaction models, similarity testing, investigation of deviations or bottleneck circumstances, and suggestion of changes (Khang et al., 2022a).

To underline the space-specific applicability of unstructured occasion log information, Martin et al. (2020) furthermore utilized AI in the fields of medical services and schooling, separately.

A top-to-bottom examination of students' web-based learning was conducted in massive open online courses (MOOCs), and a social investigation of the motivations behind why the students exited was as of late related to occasion log information that fostered a space explicit model with only a supported hypothetical model to develop esteem added administration processes.

The scholastic progress of enlisted students has been anticipated utilizing segment data, data on the training of their colleagues, grades procured in earlier coursework, and so forth. Notwithstanding, the troubles of utilizing customary information mining approaches lay in managing arrangements of occasions that consolidate dynamic ways of behaving. Choice tree strategies and bunching procedures have likewise been used to gauge students' profiles and dropout conduct.

AI is believed to have the option to overcome any barrier between data science and process science (Khang et al., 2022c).

In 2004, Luan et al. set the utilization of system mining procedures in advanced education organizations to more readily comprehend professional assistance to sign up for a particular course, and offer them help with finishing their certificate, track down a design, and conclude how previous studies played a part in proceeding to help their place of graduation in both scholar and financial vows, to give some examples.

Werner et al. (2021) took a gander at the feasibility of incorporating AI into present-day reviews and showed how solid and strong they were when contrasted with manual monetary review statements (Khanh & Khang, 2021).

Hamdan et al. (2021) showed how the closure of schools brought about by COVID-19 has impacted students from one side of the planet to the other. They likewise described the troubles that web-based guidance have introduced for teachers and students as far as learning methodology, as well as how it has assisted students with getting first-class training. Thus, students now have a greater chance to get automatic guidance and are better prepared to comprehend learning procedures while concentrating on a matter they are keen on.

An adaptable transformative tree model for process revelation, it is motivated by hereditary calculations and may improve instructive methodology, for example, in educational plan improvement, PC-assisted learning, and MOOCs.

Kas et al. (2020) showed the execution of the AutoML approach in a certifiable process mining circumstance and energized its use with successive information in additional exploration.

To represent the distinctions between employees who passed or didn't pass a given test, Azeta et al. (2020) constructed an interaction mining structure to look at the virtual learning conduct of the workforce using inductive and fluffy excavators with various wellness-level qualities.

Omori et al. (2020) introduced a thorough examination of different process devices and programming types accessible, alongside their advantages and disadvantages, Omori has shared suitable applications for testing idea float discovery situations utilizing an occasion log assembled from a modern application in view of an interaction control framework.

Kurniati et al. (2019) stressed the worth of the artificial intelligence to deal with medical services utilizing the patient's electronic well-being records (EWRs) and discussed the need to further develop information quality alongside the use of various other successful AI calculations to investigate the productivity of the ML approach in EWR systems.

According to Reimann et al. (2014), application of AI in workforce management systems is a moderately new innovation that is being created by the corporate world. It centers on the production of strategies for acquiring data about processes from occasion logs. It utilizes data framework occasion logs to find, screen, and upgrade processes across various areas, as well as to confirm process consistency, track down bottlenecks, and figure execution issues.

Most AI research has zeroed in on (business) work process frameworks and the recognizable proof of Petri network process portrayals. Occasion log information is utilized as a contribution by the methodology determined by these marks, which then, at that point, make process models that completely portray the information in the occasion logs (Gupta et al., 2023).

In our circumstance, a new area of interest is process-situated information revelation procedures in workforce management systems (Dutt et al., 2017). AI is a new and quickly growing examination idea in the field of business process management.

The essential objective of AI is to find, assess, and improve processes by getting data from a data framework's occasion logs. AI innovation requires two things: first, an idea of process working, and second, log documents that monitor each activity in the framework.

Martin et al. (2020) note that AI is being utilized in a great number of organizations (e.g., banks, medical clinics, regions, legislative offices, web shops, and cutting-edge framework producers). There are many interaction mining approaches that might answer a large number of inquiries. A view occasion is the point at which an understudy watches a specific talk. An understudy taking a test is alluded to as a test endeavor event (Bhambri et al., 2022).

14.3 METHODOLOGY

Using the search terms "(TITLE (artificial intelligence* OR AI OR workforce management OR HR OR human resource))" and "(LIMIT-TO (DOCTYPE, "article" OR "research paper")," searches in the Scopus database were conducted on December 16, 2022.

To avoid errors in the daily database updates, results were exported on the same day. No other restrictions were applied; the search was limited to the publication types "article" and "review." Articles are publications of original research, whereas reviews are publications of conference papers.

The following information was gathered: the volume of publications (total and annual), authors, and journals; the degree of open access for publications; the number of publications per journal; the journals publishing the most content and their impact factors; the language of publication; the document type; the country of publication; the affiliations of the authors; funding sponsors; the most highly cited publications; and the most published authors.

The chapters in this collection's trends were noted and identified. VOS viewer was employed to construct bibliometric networks as shown in Figure 14.1.

As a result, we are better able to use technology in our daily lives. This will make the three search terms—AI in workforce management—more clear in terms of their limitations. A closer examination of the subset of articles that a subsequent analysis of the bibliographic network has shown them to be relevant, with particular focus on the citation networks, co-occurrence networks, and the fundamental data such as nations and documents.

The first qualitative evaluation mainly focuses on the researchers' thoughts regarding the keyword selection and uses an explanatory method, whereas the bibliometric study offers more objective insights using quantitative and statistical data.

According to Fisch and Block (2020), some examples of the bibliographic data examined by bibliometric approaches include the names of the most notable authors, journal names, article titles, article keywords, and publication years.

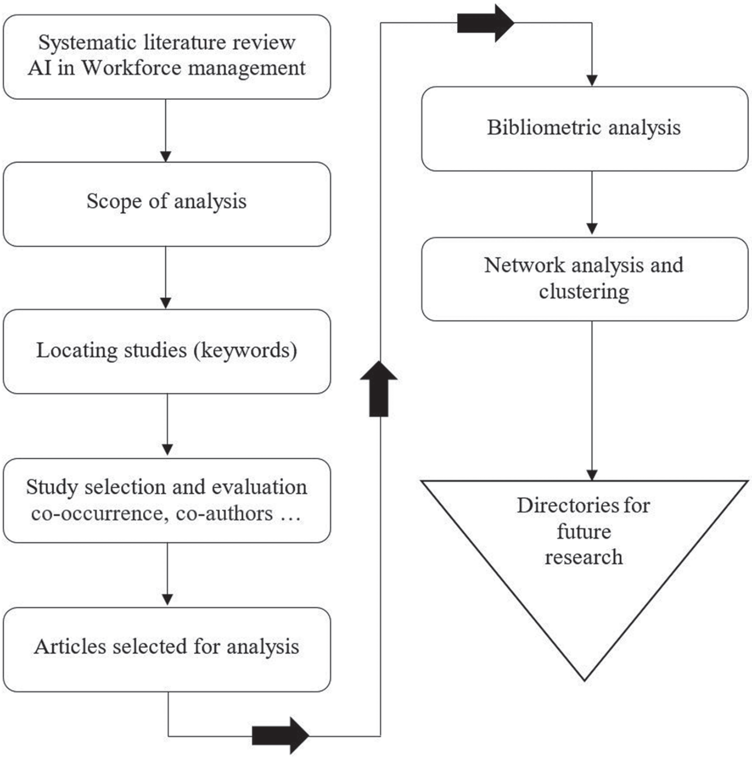

FIGURE 14.1 The flow of the study is represented in graphic format.

14.4 RELATED WORK

This study's goals are to: (1) perform a systematic literature network analysis (SLNA); (2) count the number of documents that have been published on the keywords mentioned above; and (3) identify the authors, organizations, and nations that have worked in the field of AI and workforce management.

SLNA's main objective is to analyze the field's conceptual framework and its development over time, as opposed to narrative literature evaluations, which attempt to summarize the information in the works on a specific study issue (how the have outlets evolved, etc.).

Beyond just giving a plain description of earlier works, it promotes discussion about what we already know and where we can go. By emphasizing cross-disciplinary links, it also makes it simple to keep track of how information is moved both within and between disciplines.

The avoidance of the researchers' selection bias is also made possible by quantitative bibliographic investigations, such as those by Fisch and Block (2020). As a result, the process is repeatable at any time. Last but not least, SLNA stands out because it has a broader and more current scope (e.g., in terms of the journals and publication years covered), reducing the likelihood that the authors will present an overly introspective and biased argument, instead leading to evidence-based conclusions.

Each phase of the procedure is fully described in Figure 14.1. Based on the prior research, the most pertinent articles were assessed in order to both identify the key ideas presented in each work and, more importantly, to comprehend how the topic has evolved through time. The section that follows reviews the scientific literature on the subjects of artificial intelligence and workforce management.

14.5 DATA ANALYSIS

Citation map based on bibliography data from the Scopus database files which were original under CSV format and were analyzed by using VOS viewer software. This analysis helps us to create a coauthor ship keyword co-occurrence citation bibliography coupling or citation map based on bibliography data from these Scopus database files.

SLNA was carried out using the keywords "AI" and "workforce management" from the Scopus database to find out how many chapters and who are talking about artificial intelligence. This study was undertaken to understand how, who, and where about the contributions relating to artificial intelligence in the field of workforce management.

14.5.1 Author's Network Analysis (VOS Clustering)

Coauthorship was the type of analysis used, full counting was used to count the authors, and a maximum of 25 authors per document was established. When the minimum number of documents for an author was set to 20, 14 of the 2,567 keywords reached the cutoff. Out of the 2,567 items in the network, 14 had the most related items as shown in Figure 14.2.

The node's dimension corresponds to the total link strength, and six colors—red, blue, green, yellow, pink, and orange—distinguish terms in one cluster from those in other clusters. The term clusters are looked at in the part that follows in order to talk about the most relevant research fads in the literature as Figure 14.3.

The co-occurrence network served as the study's theoretical underpinning and was utilized to focus on the authors' keywords. Co-occurrence analysis is predicated on the idea that the article keywords selected by diverse authors adequately convey the chapter's content or the connections it makes between the topics it is researching.

The goal of the analysis was to describe how the research patterns evolved over time. If a term regularly occurs in the context of other terms, this is probably an indication of a particular study pattern for the issue. a) Connors, D.P., b) Gresh, D.L., c) Naveh, Y., and d) Richter, Y. are a few academic scholars who have made major contributions to the study relating to artificial intelligence and workforce management.

14.5.2 Text Analysis (VOS Clustering)

The minimum number of occurrence of keyword was set to 5 of the 2,567 keywords that only 129 met the threshold. In order to address the most prevalent terms in the literature and identify potential research avenues within each cluster, the following

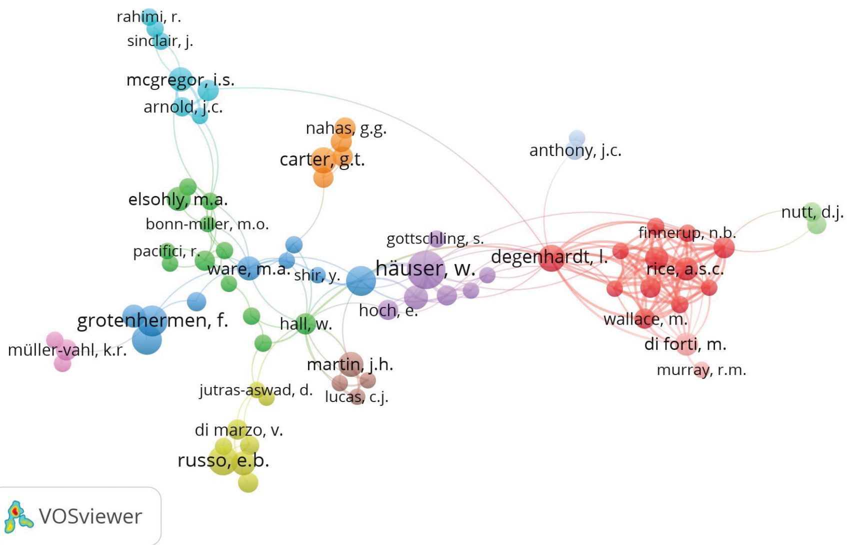

FIGURE 14.2 Visualization of clusters that represents the link strength between the coauthors with regard to co-occurrence and relativity in research.

***Source*: Author's own analysis from VOSviewer 1.6.18.**

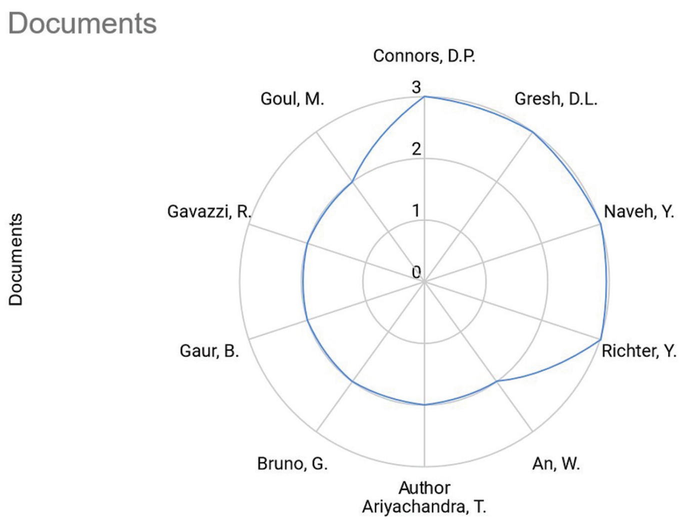

FIGURE 14.3 Radar chart of the documents of the top ten authors.

***Source*: Author's own analysis from VOSviewer 1.6.18.**

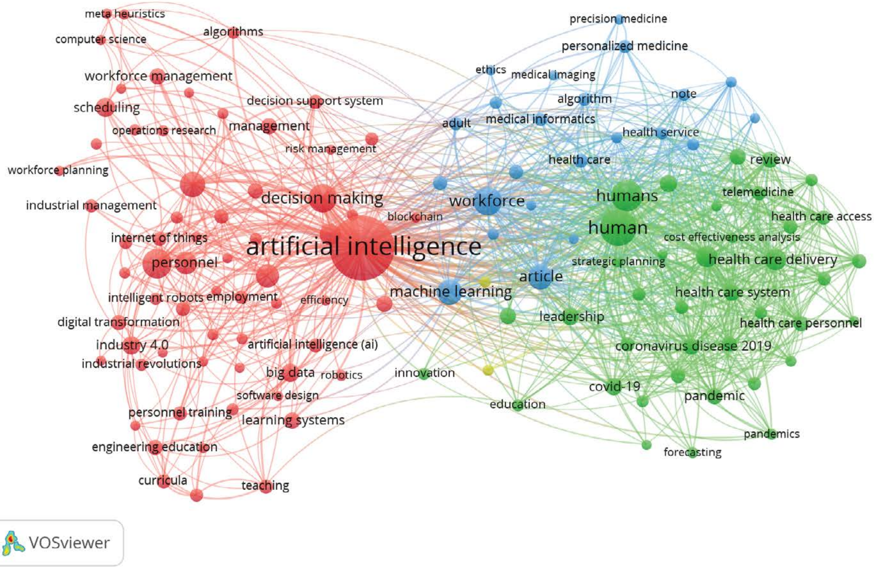

FIGURE 14.4 Co-occurrence author keywords network.

***Source*: Author's own analysis from VOSviewer 1.6.18.**

concerns have been examined using the results of a VOS viewer visualization as Figure 14.4.

As most occurred items in the below Figure 14.5, the red-gray cluster shows up for AI, decision-making, digital transformation, ML, personnel training, workforce management, etc. The blue-gray cluster shows up for algorithm, machine learning, healthcare, health services, etc. There is a huge amount of evidence of work and availability of literature regarding AI and its application in the areas of healthcare and education.

As most occurred items in Figure 14.6, the red-gray cluster shows up for innovation, opportunity, technology, health, implementation, etc. The blue-gray cluster shows up for technique, operation, problem, worker, machine, etc. The purple-gray cluster shows up for student effectiveness, education, learning, etc. The map was made using text data, or "term co-occurrence" data.

Using CSV files from the Scopus database, the data was read from the bibliography database files. The abstract fields of the articles were discarded, but the title fields of those articles were extracted. Binary counting was rejected in favor of full counting. Only 113 of the 7,217 terms met the requirement for having a minimum number of ten occurrences of the same phrase.

The conclusion drawn out from the above cluster could be that a large number of studies were conducted in the field of medicine and education.

The green cluster shows innovation, implementation, healthcare, opportunity, etc., as the most frequently occurring items. The blue cluster shows technique, operation, problem, worker, machine, etc. The purple cluster shows student effectiveness,

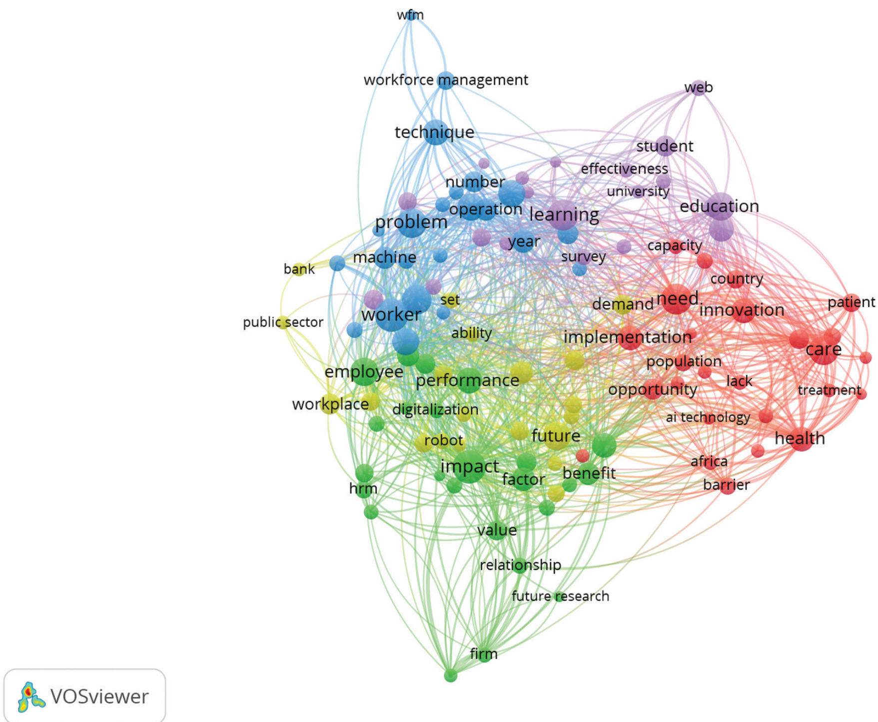

FIGURE 14.5 Co-occurrence author keywords network in the areas of healthcare and education.

***Source*: Author's own analysis from VOSviewer 1.6.18 (Nees Jan van Eck and Ludo Waltman).**

education, learning, etc. The map was made using text data, or "term co-occurrence" data. Using CSV files from the Scopus database, the data was read from the bibliography database files (Rani et al., 2021).

The abstract fields of the articles were discarded, but the title fields of those articles were extracted. Binary counting was rejected in favor of full counting. The map as Figure 14.7 was made using text data, or "term co-occurrence" data. In order to maintain the default percentage setting of 60% in the VOS viewer software, 51 terms were chosen out of a total of 2,567 terms when the calculation was done to meet the requirement for 20 or more occurrences of the same phrase. The cluster is constituted with a total of 51 terms which are interconnected the color determines them is one factor or belong to family or group.

The conclusion drawn out from the above cluster could be that a large number of studies were conducted in the field of medicine and education.

The top 14 pertinent keywords were compiled and divided into three groups, as seen in Figure 14.8. The 2,567 keywords are the network's nodes, and each phrase's